

Enterprise Learning & Workforce Management Platform

Milestone 3 – Progress Analytics, Coding Practice, Contests & Career Roadmap

My Progress · Coding Practice · Contests · Career Roadmap

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1. Objective

With the core learning loop from Milestone 2 in place — enroll, learn, assess, certify — Milestone 3 focuses on depth: giving students a detailed view of their own learning velocity, a space to practice technical skills independently of any single course, a competitive environment to test those skills against peers, and a structured, role-based roadmap that ties everything together into a career plan.

The main objectives were:

- Build the full My Progress analytics page, going beyond the dashboard summary from Milestone 2.
- Implement a standalone Coding Practice module with an in-browser editor and automated evaluation.
- Implement time-boxed Contests with registration, live leaderboards, and contest certificates.
- Implement a role-based Career Roadmap with a week-by-week curriculum and daily study plans.
- Design the supporting database tables for practice, contests, and roadmap content.

2. My Progress

This is the Learning Velocity & Performance page — the detailed counterpart to the lightweight dashboard summary introduced in Milestone 2.

2.1 Stat Cards

Four stat cards summarise a student's overall learning activity, all computed from their enrollments records:

- Total Enrolled — number of courses the student has joined
- Average Completion % — mean progress across all enrollments
- Courses Completed — count of enrollments at 100% progress
- Active In-Progress — count of enrollments between 0% and 100%

2.2 Charts

Two charts are rendered using Recharts: an overall completion chart summarising the student's progress as a whole, and a course-wise chart that compares completion percentage across each individually enrolled course, helping a student spot which course needs more attention.

2.3 Data Flow

```

My Progress page loads
↓
Axios GET /api/dashboard/progress
↓
DashboardController → aggregation query over enrollments
↓
Compute: total, avg%, completed, in-progress
↓
Return stats + per-course progress array
↓
Recharts renders overall + course-wise charts

```

If no courses are enrolled, an empty state is shown instead of an empty chart; if the request fails (e.g. a network error), a retry action is shown instead of leaving the page stuck loading.

3. Coding Practice

Coding Practice is a standalone skill-building area, independent of any specific enrolled course, aimed at general programming proficiency.

3.1 Practice Flow

```

Student opens a Coding Question
↓
In-browser code editor loads
↓
Student writes solution & submits
↓
CodingSubmission created
↓
Run against CodingTestCases
↓
Pass/Fail result recorded
↓
CodingProgress updated (solved count, streak)
↓
CodingLeaderboard re-ranked

```

3.2 Difficulty Tiers, Badges & Leaderboard

Questions are organised by difficulty (Easy / Medium / Hard) so students can build up gradually. CodingBadges are awarded for reaching milestones (e.g. a solve-count threshold or a practice streak), and a global CodingLeaderboard ranks students platform-wide, adding a light competitive incentive to regular practice even outside of formal contests.

4. Contests

Contests layer time-boxed, competitive practice on top of the always-available Coding Practice module.

4.1 Contest Lifecycle

```

Contest published (admin)
↓
Student registers → ContestRegistration
↓
Contest window opens
↓
Student solves ContestQuestions
↓
Submission run against ContestTestCases → ContestSubmission
↓
ContestResults calculated
↓
ContestLeaderboard updated live
↓
Contest window closes
↓
Top performers → ContestCertificate auto-issued

```

4.2 Discussion & Recognition

Each contest has its own ContestDiscussion thread so participants can talk about problems after the contest ends, and ContestBadges recognise strong or consistent performance across contests. Contest certificates reuse the same certificateId / QR / public-verification pattern built for course certificates in Milestone 2, so competitive achievement is just as verifiable as course completion.

5. Career Roadmap

The Career Roadmap is the platform's role-based career-preparation system, tying course learning, coding practice, projects, and certifications into one structured plan per career track.

5.1 Career Tracks

Roadmaps are currently defined for five tracks:

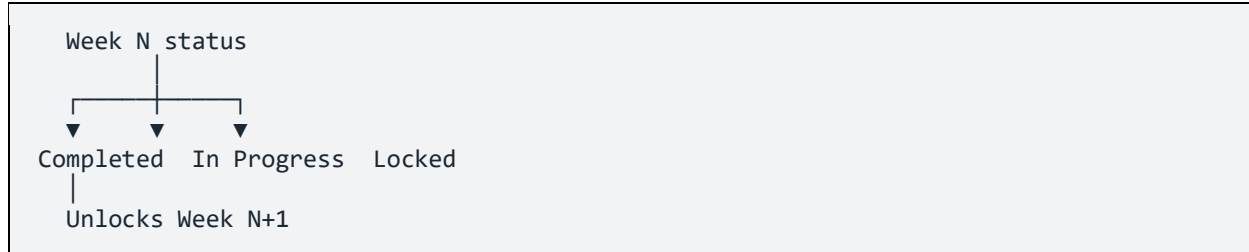
- Full Stack Developer
- Java Backend Developer
- Frontend Developer
- Python Developer

- Cloud Engineer

Each CareerRoadmap entity stores a unique role key, title, description, and estimated duration, along with JSON-encoded lists of required skills, recommended courses, recommended certifications, coding-practice topics, projects, and interview-preparation content specific to that role.

5.2 Weekly Curriculum & Submission-Based Progression

Within a chosen track, the roadmap is organised as a 12-week semester curriculum. Each week lists its topics, estimated hours, assignment submission, mini-project submission, and overall progress, and unlocks sequentially based on submitted work:



A 'View Daily Plan' action drills into a detailed 7-day study and coding schedule for that week — daily learning goals, time slots, tutorials, and a daily coding challenge — so the broad weekly plan is always backed by a concrete, actionable day-by-day routine.

6. Core Backend Classes

DashboardController.java

Aggregates enrollment and lesson-progress data into the stat-card and chart figures shown on the My Progress page.

CodingPracticeController.java

Serves coding questions and test cases, accepts submissions, and returns progress, badge, and leaderboard data.

ContestController.java

Handles contest listing, registration, submission, live leaderboard queries, and contest-certificate issuance.

CareerRoadmapController.java

Serves roadmap definitions per role key, including the weekly curriculum structure used by the frontend to render status and the 7-day daily plan.

7. Database Design for This Milestone

Coding Practice tables:

- coding_languages
- coding_questions
- coding_test_cases
- coding_submissions
- coding_progress

- coding_badges
- coding_leaderboard

Contest Platform tables:

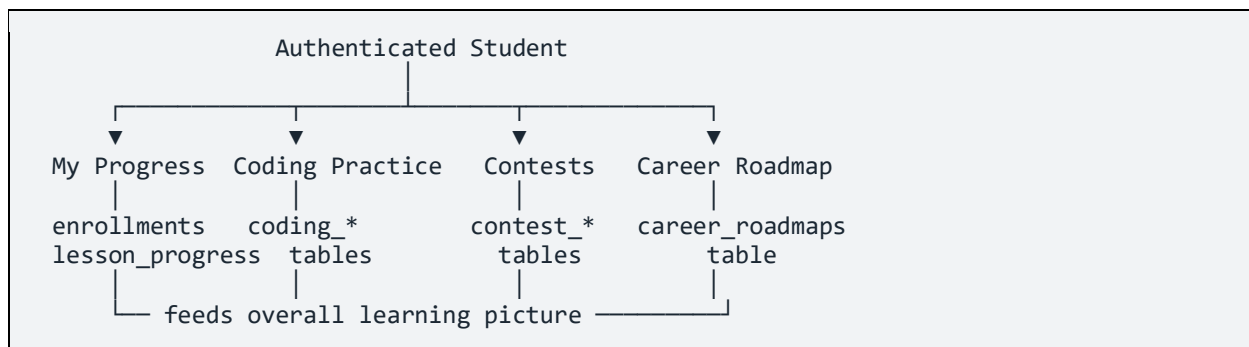
- contests
- contest_registrations
- contest_questions
- contest_test_cases
- contest_submissions
- contest_results
- contest_leaderboard
- contest_certificates
- contest_badges
- contest_notifications
- contest_discussions

Career Roadmap table:

- career_roadmaps — role_key, title, description, estimated_duration, and JSON text columns for skills / courses / certifications / coding topics / projects / interview prep

My Progress does not introduce new tables — it is a read-side aggregation over the enrollments and lesson_progress tables already built in Milestone 2.

8. Complete Milestone 3 Architecture



9. Conclusion

Milestone 3 deepened the platform from a linear course-completion flow into a fuller learning ecosystem. Students can now see exactly how they're progressing, sharpen their skills independently through Coding Practice, test themselves competitively in Contests, and follow a structured, role-based Career Roadmap that ties learning, practice, and certification into a single job-ready plan.

With progress, practice, and career planning in place, the platform is ready for Milestone 4, which adds the remaining career-readiness tools, platform utilities, and admin capabilities needed to complete the student and administrator experience end to end.